

An Empirical Study on Utilizing Large Language Models for Bengali Image Caption Generation

Muhammad Azmain Mahtab, Jannatim Maisha, Md. Masudur Rahman, Sajib Kumar Saha Joy
Department of Computer Science and Engineering
Ahsanullah University of Science and Technology
Dhaka, Bangladesh
azmainmah@gmail.com, maishapousy@gmail.com, masudur.rahman0413.cse@aust.edu, joy.cse@aust.edu

Abstract—An exemplary caption not only describes what is happening in a particular image but also denotes intricate traditional objects in the image by their local representative terms through which the native speakers can recognize the object in question. A caption that fails to accomplish the latter is not effective in conveying proper utility. To ensure caption locality, we aim to explore the potential of Large Language Models (LLMs) in Bengali image captioning, which have lately shown promising results in English language caption generation. As a first for the Bengali language, we utilized CLIP (Contrastive Language-Image Pre-training) encodings as a prefix to the captions by employing a mapping network, followed by fine-tuning BanglaGPT, a Bengali pre-trained large language model to generate the image captions. Furthermore, we explored vision transformer-based encoders (ViT, Swin) with BanglaGPT as the decoder. The best BanglaGPT-based model outperformed the current benchmark results, with BLEU-4, METEOR, and CIDEr scores of 54.3, 39.2, and 95.9 on the BanglaLekha dataset and 67.4, 36.6, and 76.9 on the BNature dataset.

Index Terms—Bengali image captioning, Large Language Model, CLIP, Vision Transformer, GPT-2, BanglaGPT

I. INTRODUCTION

Image captioning, a multi-modal task of generating natural language descriptions from images, provides critical assistance to people with visual impairments by translating visual content into descriptive text. This enables people to view and interact with digital content ranging from social media posts to educational resources, enabling more inclusion and equitable opportunities in today's environment. Substantial progress has been made in image captioning in several languages. Despite being the world's sixth¹ most spoken native language and seventh² in terms of total speakers, Bengali has received inadequate attention in this sector. This language, rich in cultural nuances, poses particular issues for image captioning due to the possibility of misinterpretation of cultural references or subtleties within the imagery. One potential approach could be translating captions generated by state-of-the-art English language-trained models into Bengali. However, this strategy is likely to miss the intricacies and cultural references found in Bengali imagery. As shown in Figure 1, a state-of-the-art model [1] trained on the COCO dataset fails to identify Bengali cultural icons like rickshaws. In contrast, a model trained



COCO Model: A couple of men riding on the back of a horse drawn carriage.
Our Bangla Trained Model: রিকশা চালক রাস্তায় গাড়ি চালাচ্ছে।
(The rickshaw puller is driving down the road.)

Fig. 1. Comparison between English dataset trained model and our Bengali dataset trained model generated captions.

on a native Bengali dataset can easily identify these objects. Therefore, simply translating English captions to Bengali does not produce the desired results.

Current research on Bengali image captioning combines CNN-based encoders with RNN-based decoders. Transformer-based encoders/decoders have also been implemented. However, none have used contrastive objective pre-trained visual encoders or generative LLMs as decoders.

To accomplish our goal of Bengali image captioning while addressing the research gap, we experimented with a pair of methods. The first method uses CLIP (Contrastive Language-Image Pre-training) [2] to create a semantic prefix for the caption, which is then fed into a large language model. This technique is proven effective in the work described in [1]. The second approach uses a conventional encoder-decoder architecture. A vision transformer will be utilized as the image encoder, and an LLM as the text decoder, similar to the approach described in [3].

To summarize the main contributions of our work:

- We leverage the state-of-the-art CLIP model for extracting visual features and an autoregressive language model for text generation. This is the first time these models

¹ wikipedia.org/wiki/List_of_languages_by_number_of_native_speakers

² wikipedia.org/wiki/List_of_languages_by_total_number_of_speakers

have been utilized for Bengali image captioning.

- To gain a better grasp of Bengali LLM caption generation, we conduct prefix and beam size analyses.
- The GPT-2 based models achieved benchmark performance on BanglaLekhaImageCaptions and BNATURE datasets, significantly outperforming traditional transformer-decoder-based architectures.

II. RELATED WORKS

RNN-based Encoder-decoder Scheme. [4] introduces “Chitron,” an automatic Bangla image captioning system using VGG16 and stacked LSTM, along with the “BanglaLekha-ImageCaptions” dataset containing 16,000 culturally relevant images with Bangla captions. [5] presents “BanglaLekhaImageCaptions” dataset containing 9,154 images with two Bangla annotations each, spanning 25 classes, focusing on Bangla imagery. They combined a VGG-16 with LSTM for Bangla caption generation. [6] proposes a CNN (InceptionV3) and Bi-GRU-based encoder-decoder model for generating Bangla captions. They create a new Bengali dataset called BNATURE, containing 8,000 images with five captions each. [7] combined fastText and GloVe word embeddings and created a hybridized encoder-decoder based approach.

Attention-based RNN Encoder-decoder Scheme. [8] proposes a InceptionV3 encoder and a GRU-based decoder with attention, incorporating a novel joint loss function combining encoder attention loss and decoder sequence loss. [9] proposes a hybridized attention-based model, combining an InceptionV3 encoder with visual attention and a Bidirectional GRU based decoder with attention.

Transformer-based Encoder-decoder Scheme. [10] utilizes a pre-trained ResNet-101 encoder and a Transformer decoder with multi-head attention and cross-attention, a first for Bengali image captioning. Word embeddings and positional encoding are derived from the sahaJBERT model. [11] compared an InceptionV3 and transformer based encoder-decoder model to a visual attention based model (InceptionV3/Xception encoders and GRU decoders with visual attention). They introduced the “Bornon” dataset containing 4100 images with 5 captions each. [12] annotated 158,915 Bangla descriptions for the Flickr30k images, naming the dataset “Biboron”. They compared a transformer encoder-decoder model with multi-head attention to a VGG-16 and Bi-LSTM based model.

CNN Encoder-decoder Scheme. In [13], the authors argue that CNNs capture fine-grained sentence structure and linguistic diversity better than RNNs. They combine a ResNet-50 CNN based encoder with a 1D-CNN with word embeddings for generating linguistic representations.

III. METHODS

A. CLIP-Encoder Prefix Based

In this method, we leverage the CLIP (Contrastive Language-Image Pretraining) model, a pre-trained neural network adept at generating semantic encodings for various images without additional supervision. We complement CLIP

with fine-tuned Generative Pre-trained Transformer 2 (GPT-2) in order to create meaningful and contextually relevant image captions in Bengali.

CLIP Encoding. Initially, we obtain image embeddings using the CLIP model’s image encoders. Two architectures, Vision Transformer (ViT-L/14) and Residual Networks (ResNet50x4), are employed as image encoders. These encoders generate image embeddings, with ViT-L/14 embeddings having a shape of [total captions, 768] and ResNet50x4 embeddings a shape of [total captions, 640].

LLM Fine-tuning. Image embeddings are processed by a Multi-Layer Perceptron (MLP), generating a sentence of predefined length (prefix length) ranging from 6 to 12. Subsequently, the generated tokens are concatenated with ground truth textual captions. The combined list of tokens, comprising both image and caption tokens, is utilized for fine-tuning BanglaGPT [14], a Bengali pre-trained generative large language model. BanglaGPT consists of a maximum context length of 512 compared to 1024 for the base GPT-2 variant. It has 124M parameters along with a Beginning-of-Sequence Token ID of 0 and an End-of-Sequence Token ID of 2.

Caption Generation. In the caption generation phase, Beam Search [15] is employed, taking the model, BanglaGPT tokenizer, and prefix embeddings as parameters. We adopt a simple Multi-Layer Perceptron as the Mapping Network, utilizing a single hidden layer given CLIP’s pre-training for vision-language tasks. The MLP projects image features into the same space as the BanglaGPT embeddings, facilitating the generation of fixed-length prefix text from image embeddings. During fine-tuning, the MLP network remains frozen, while the language model undergoes training.

Beam Search. Text generation employs beam search, a systematic exploration strategy that evaluates multiple word sequences, or “beams,” based on a scoring mechanism. At each step, the model generates logits for the next token, which are normalized using the Softmax activation function and adjusted by the Temperature parameter to introduce randomness.

Initially, the top beam-size tokens are selected as candidates for each beam. Subsequent steps involve selecting top candidates based on the sum of scores along each beam, ensuring that stopped beams are not considered. Selected tokens are appended to the sequences and utilized for the next step of generation. In Figure 2, caption generation is visualized given a beam size of 2.

B. Vision Transformer Encoder Based

The Vision Encoder-Decoder model bridges the gap between image understanding and text generation. It leverages pre-trained vision transformer models, ViT [16] and Swin [17], for image encoding, and BanglaGPT, a Bengali language pre-trained GPT-2, for text sequence generation.

Image Encoder. We utilize two pre-trained vision transformer models for image encoding. For ViT, we employ the base variant³ with a patch size of 16x16 pixels and an input

³huggingface.co/google/vit-base-patch16-224-in21k

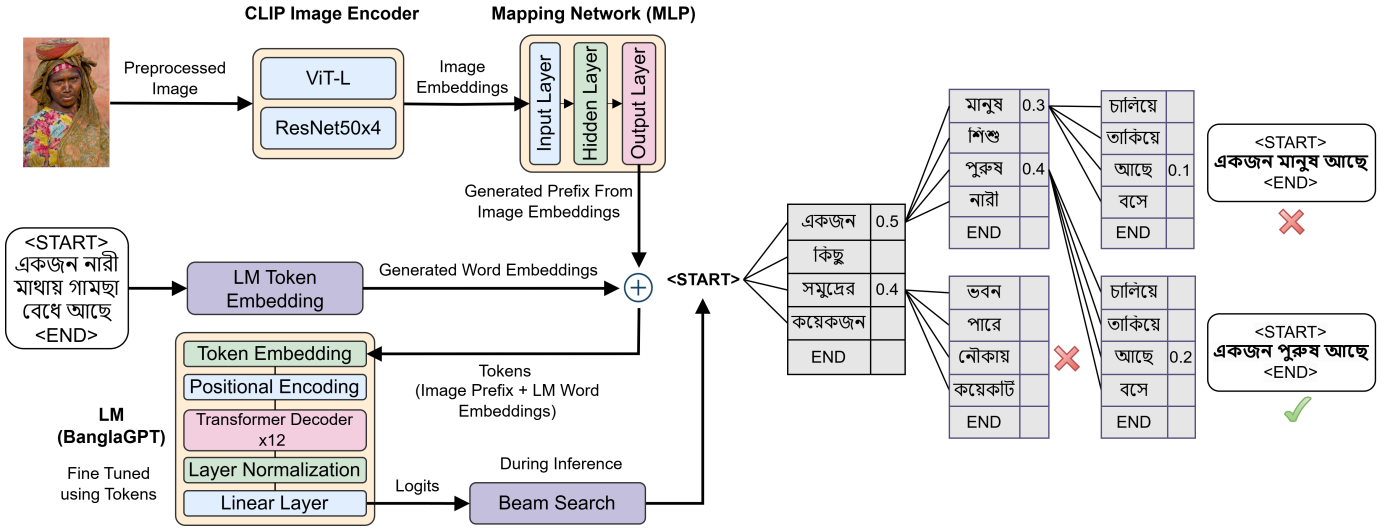


Fig. 2. CLIP Encoder based method.

image size of 224x224, pre-trained on ImageNet-21K. For Swin, we adopted a base variant⁴ with a patch size of 4x4, a window size of 12, and an input image size of 384x384, pre-trained on ImageNet-21K.

Text Sequence Decoder. BanglaGPT with 12 decoder blocks, serves as the text decoder.

Cross-Attention Layer. This layer connects the image and text embeddings, enabling the model to incorporate both visual and textual information during caption generation. The image encoder's embeddings are used as Key (K) and Value (V), while the decoder's embeddings are used as Query (Q) in the cross-attention head.

Training and Generation. During training, the model learns to produce the input caption based on the image and text inputs. The model's output layer produces a vector of size [Sequence Length, Vocabulary Size]. After applying Softmax function, the final sequence is generated. This generated sequence is then compared to the input caption, and a loss function calculates the penalty for incorrect words. Backpropagation minimizes the calculated loss, allowing the model to learn from its errors. Following training, caption generation proceeds as all words except the one following the start token are masked. The model progressively predicts the masked words one by one, continuing until it encounters the end token or reaches the maximum output sequence length.

IV. EXPERIMENTS

A. Dataset and Evaluation Metrics

We evaluate the models on two native Bengali image captioning datasets: BanglaLekhaImageCaptions (9,154 images with 2 captions each with a total of 18,308 captions), BNATURE (8,000 images with 5 captions each with a total of 40,000 captions). Details of the dataset distributions can be found in Table I. We employed conventional image caption

evaluation metrics, i.e., CIDEr [18], METEOR [19], and BLEU [20] to evaluate the generated captions. We applied Unicode normalization using the NFC (Normalization Form C) method on captions before evaluation.

TABLE I
DATASET SPLITTING RATIO

Dataset	Total	Training	Validation	Testing
BanglaLekha ImageCaptions	9154	7414(81%)	824(9%)	916(10%)
BNATURE	8000	6480(81%)	640(8%)	880(11%)

B. Experimental Setup

CLIP Encoder Based model has been trained with batch size variations of 8, 16, 32, and 40. The prefix length for the image encodings after passing through the MLP has been set to 10. We constructed MLP using a single hidden layer and with Tanh activation function. Furthermore, this model was trained for 6 and 10 epochs using the AdamW Optimizer with a learning rate of 2×10^{-5} and a custom learning rate scheduler, with 5000 as the warmup step. As loss function, the Cross Entropy loss was used. For caption generation, beam size was set to 3, 4, 5, 6, and the max length of the sequence was set to 120 along with temperature set to 1.

Vision Transformer Encoder Based model has been trained with batch size variations of 8, 16. Furthermore, this model was trained for 5, 10, 15, and 20 epochs using AdamW Optimizer with a learning rate of 5×10^{-5} and a custom learning rate scheduler, with 1024 as the warmup step. As loss function, the Cross Entropy loss was used. The caption generation was done using Beam Search technique where the beam size was set to 2 and the max length of the sequence was set to 64. The size of n-grams that should not be repeated in the generated sequences, is set to 3, and the length penalty applied during beam search decoding is set to 2.

⁴huggingface.co/microsoft/swin-base-patch4-window12-384-in22k

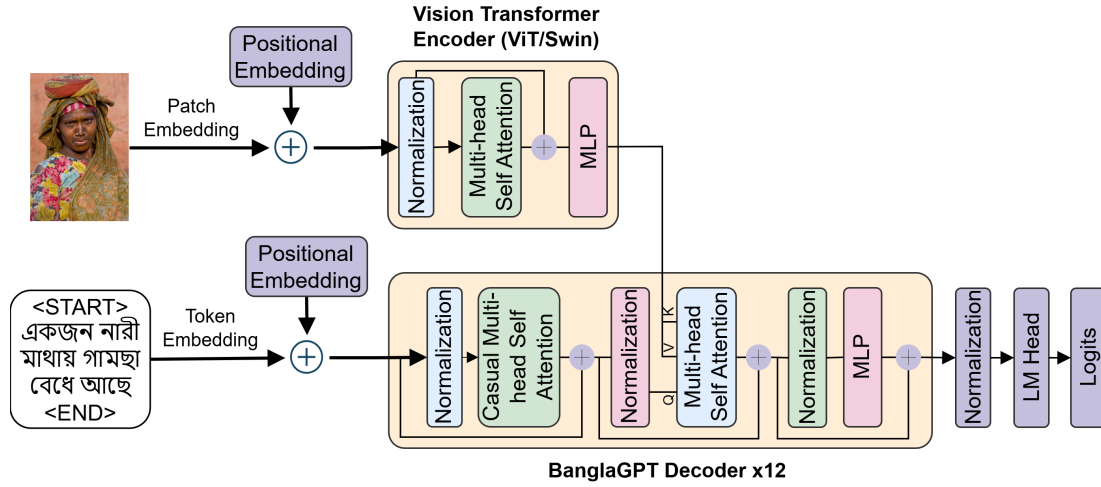


Fig. 3. Vision Transformer Encoder based method.

Implementation Details. Model training was done on PyTorch 2.1.1 with an NVIDIA RTX 3070 8GB GPU. The clip-based models took one to five hours to train and the vision encoder-decoder models took one to seven hours depending on hyperparameters.

Baseline models. We evaluated the performance of our methods against several recent benchmark Bengali image captioning studies, including [5], [6], [8], [9], [11], [13]. Note that the dataset splits may vary as none seem to follow a standard split.

C. Result Analysis

Batch Size and Training Epoch. In Table II, we observed a trend of decreasing performance with increasing batch size. A batch size of 8 yielded the best results for both encoders due to the small number of data. Similar patterns were observed in Table III for the BNATURE dataset. Notably, the ViT-L encoder achieved slightly superior results compared to the ResNet50x4 encoder. The difference in performance can be attributed to the fact that the more complex ViT-L architecture (304M parameters) was able to capture more image features compared to ResNet50x4 (87M parameters). We also tested the ViT-B (88M parameters) encoder, which produced low scores compared to ViT-L and ResNet50x4. The optimal number of training epochs differed between encoders (10 for ResNet50x4 vs 6 for ViT-L).

Prefix Length. As seen in [21], extending the length of the prefix up to a certain point enhances the model's performance in an underlying task. Furthermore, the saturation length may vary between operations. Considering our observations in Table IV, we can point out that the MLP architecture is intrinsically constrained because it cannot scale for lengthy prefixes when training for small number of epochs. But with increased 10 epochs of training, larger lengths produced the best result. Note that we are talking about small margins here.

Beam Size. We evaluated the impact of beam size on the performance of the CLIP-based model using the BanglaLekha

TABLE II
METRICS FOR BANGLALEKHA IMAGE CAPTIONS DATASET ON CLIP BASED MODELS

Encoder	Batch Size	Epoch	B@1	B@2	B@3	B@4	M.	C.
ResNet50x4	32	10	65.6	60.0	55.4	51.1	37.0	84.6
		6	60.9	55.5	51.1	47.0	33.0	70.5
	16	10	67.9	61.8	56.9	52.5	38.2	90.5
		6	66.2	60.2	55.6	51.4	36.6	90.8
	8	10	68.5	62.2	57.2	52.7	37.1	88.0
		6	66.3	60.5	55.9	51.6	38.0	93.5
ViT-L/14	32	10	65.0	59.6	55.1	51.0	37.1	88.1
		6	58.2	53.7	49.9	46.2	34.6	80.2
	16	10	68.6	62.3	57.3	52.9	38.1	91.6
		6	67.1	61.4	56.7	52.4	37.1	88.3
	8	10	68.5	62.1	57.0	52.5	37.6	86.4
		6	68.6	62.7	57.8	53.4	38.7	92.3

TABLE III
METRICS FOR BNATURE DATASET ON CLIP BASED MODEL

Encoder	Batch Size	Epoch	B@1	B@2	B@3	B@4	M.	C.
ResNet50x4	16	10	82.4	75.7	69.8	64.6	33.4	65.8
		6	80.9	75.3	70.3	65.9	33.5	63.8
	8	10	82.5	76.1	70.5	65.5	34.1	69.5
		6	82.3	76.0	70.6	65.8	34.7	69.4
ViT-L/14	16	10	82.8	76.4	71.0	66.2	34.3	71.7
		6	80.9	75.2	70.2	65.8	33.9	66.4
	8	10	81.4	75.2	69.8	64.9	33.6	68.0
		6	82.4	76.8	71.9	67.4	36.6	76.9

dataset (Table V). Beam size controls the number of candidate captions considered during decoding. While a larger beam size can lead to more diverse captions, it might also introduce inaccuracies. Our experiments revealed a trade-off between caption accuracy and informativeness. A beam size of 3 resulted in the highest BLEU-1 score (69.0), indicating better performance for short and essential phrases, but the captions generated were found to be conservative. Conversely, a beam size of 6 achieved the best CIDEr (92.4) and METEOR (38.8) scores, suggesting a good balance between informativeness and accuracy. However, the overall performance difference between varying beam size were negligible, with no major

TABLE IV
METRICS FOR DIFFERENT PREFIX LENGTHS ON BEST REPORTED CLIP
BASED MODEL

Epoch	Prefix Length	B@1	B@2	B@3	B@4	M.	C.
6	6	69.9	63.6	58.5	53.9	38.4	92.2
	8	67.7	61.9	57.2	53.0	38.2	94.7
	10	68.6	62.7	57.8	53.4	38.7	92.3
	12	67.5	61.7	57.0	52.8	38.6	92.1
10	6	68.9	62.4	57.2	52.6	37.8	86.7
	8	68.9	62.6	57.7	53.3	39.1	92.8
	10	68.5	62.1	57.0	52.5	37.6	86.4
	12	70.2	63.9	58.8	54.3	39.2	95.9

TABLE V
METRICS FOR DIFFERENT BEAM SIZES ON BEST REPORTED CLIP BASED
MODEL

Beam Size	B@1	B@2	B@3	B@4	M.	C.
3	69.0	62.9	57.9	53.3	38.4	90.7
4	68.9	62.9	58.0	53.5	38.7	91.8
5	68.6	62.7	57.8	53.4	38.7	92.3
6	68.6	62.7	57.9	53.4	38.8	92.4

improvement going above beam size of 5.

Validation. As shown in Figure 4, the training process for the CLIP-based model, using the BanglaLekha dataset for 10 epochs with a batch size of 32, suggest potential overfitting nearing 10 epochs, evidenced by the plateauing of validation accuracy and the subsequent increase in validation loss.

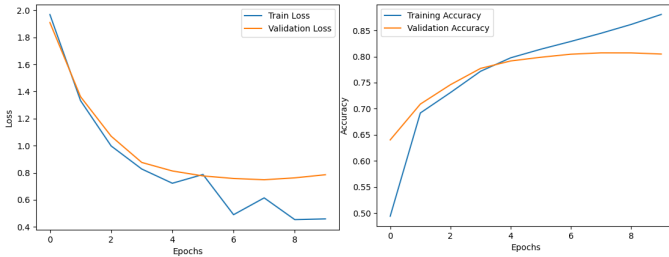


Fig. 4. Loss (left) and Accuracy (right) plots for CLIP-ViT.

As seen in Tables VI and VII, the more complex Swin-B encoder outperformed the ViT-B encoder for both datasets using transformer encoder-decoder based models. However, for BanglaLekhaImageCaptions dataset, a higher batch size of 16 showed better performance in contrast to the CLIP based models.

D. Comparisons with Baseline Models

As evident in Table VIII, on the BanglaLekha dataset, the CLIP-based model outperformed methods from previous studies. Compared to the current benchmark metrics, our configuration yielded 15.2% higher BLEU-3 scores, 28.4% higher BLEU-4 scores, 15.1% higher METEOR scores and 62.7% higher CIDEr scores. Similarly, using the BNATURE dataset, the CLIP-Based model outperformed previous models by a large margin, with 31.7% higher BLEU-1 scores, 42.1% higher BLEU-2 scores, 48.8% higher BLEU-3 scores, 53.9% higher BLEU-4 scores, and 24.2% higher METEOR scores. Comparison between the clip and transformer based approaches

TABLE VI
METRICS FOR BANGLALEKHAIMAGECAPTIONS DATASET ON
TRANSFORMER ENCODER BASED MODELS

Encoder	Batch Size	Epoch	B@1	B@2	B@3	B@4	M.	C.
ViT-B	8	5	70.7	62.4	56.0	50.6	33.2	69.8
		10	70.6	61.1	53.8	47.8	29.6	55.4
		15	70.9	61.5	54.4	48.5	30.9	63.2
		20	71.3	62.3	55.2	49.4	32.0	64.0
	16	5	69.6	61.3	54.9	49.4	33.2	65.3
		10	70.9	61.7	54.5	48.5	30.3	55.6
Swin-B	8	5	72.4	64.3	57.9	52.6	35.4	77.9
		10	72.5	63.7	57.0	51.3	34.0	68.2
		15	72.8	64.3	57.7	52.1	34.0	74.2
		20	72.7	64.3	57.8	52.2	34.3	71.6
	16	5	72.0	64.1	57.9	52.7	36.4	79.5
		10	72.6	64.4	58.0	52.5	35.1	74.7

TABLE VII
METRICS FOR BNATURE DATASET ON TRANSFORMER ENCODER BASED
MODELS

Encoder	Batch Size	Epoch	B@1	B@2	B@3	B@4	M.	C.
ViT-B	8	5	82.3	74.2	67.3	61.4	29.7	55.3
	16	5	83.0	75.4	68.9	63.2	30.9	57.6
Swin-B	8	5	83.4	76.1	69.8	64.3	32.4	60.8
		10	83.0	75.5	69.2	63.8	32.4	68.0
	16	5	82.3	75.2	69.1	63.7	32.4	63.0
		10	82.6	75.4	69.1	63.5	31.9	62.1



Clip-ResNet: একটি মেয়ে শিশু তাকিয়ে আছে।
(A young girl is looking at the camera.) **M-91.4**
Clip-ViT: একটি মেয়ে শিশু তাকিয়ে আছে।
(A young girl is looking at the camera.) **M-91.4**
Swin-B: একটি মেয়ে শিশু তাকিয়ে আছে।
(A young girl is looking at the camera.) **M-91.4**
ViT-B: একটি শিশু তাকিয়ে আছে।
(A child is looking at the camera.) **M-99.2**



Clip-ResNet: ফসলের ক্ষেতে একজন নারী আছে।
(There is a woman in the harvest field.) **M-8.5**
Clip-ViT: একজন নারী আছে।
(There is a woman.) **M-8.8**
Swin-B: একজন নারী ফুল নিয়ে আছে।
(A woman is holding flowers.) **M-21.8**
ViT-B: একজন নারী একটি শিশু কোলে নিয়ে আছে।
(A woman is holding a child.) **M-11.4**

Fig. 5. Visualization of Bengali captions generated by best reported CLIP and Vision Transformer based models on BanglaLekha test samples. Note that colored fonts denote Meteor scores.

revealed that both are quite similar in performance, but CLIP encoder is proven more effective in capturing visual features. The significant improvement in higher n-gram BLEU scores (BLEU-3 and BLEU-4) indicates that the models, powered by BanglaGPT, excel at producing longer sentences that are more contextually and coherently similar to the reference captions. This demonstrates their ability to capture long, particular sequences. Furthermore, the high METEOR scores indicate excellent performance in detecting semantic similarity between generated and ground truth captions. METEOR takes into account synonyms and other word forms in addition to exact word matches. This exhibits the models' ability to properly capture the text's meaning and structure.

TABLE VIII
COMPARISON OF BLEU, METEOR, AND CIDER SCORES WITH EXISTING BASELINE MODELS.

Dataset	Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	CIDEr
BanglaLekha ImageCaptions	CNN-ResNet-50 [13]	65.1	42.6	27.8	17.5	29.7	50.1
	VGG-16 + LSTM [5]	66.7	43.6	31.5	23.8	18.2	-
	InceptionV3 + GRU [8]	67	-	-	26	-	-
	Xception + BiGRU [7]	67.4	52.7	45.4	34.4	-	-
	InceptionV3 + Transformer [11]	66.5	55.6	47.6	40.8	25.5	-
	ResNet-101 + Transformer [10]	69.4	58.0	50.5	-	33.7	-
	Clip-ViT + BanglaGPT	70.2	63.9	58.8	54.3	39.2	95.9
	SwinB + BanglaGPT	72.0	64.1	57.9	52.7	36.4	79.5
BNATURE	InceptionV3 + BiGRU [6]	42.58	27.95	23.66	16.41	28.70	-
	InceptionV3 + BiGRU [9]	60.6	50.1	43.7	38.8	-	-
	Clip-ViT + BanglaGPT	82.4	76.8	71.9	67.4	36.6	76.9
	SwinB + BanglaGPT	83.4	76.1	69.8	64.3	32.4	60.8

V. CONCLUSION AND FUTURE WORK

The BanglaGPT-based models significantly outperform current benchmarks on publicly available Bengali image captioning datasets, highlighting the potential of pre-trained generative large language models for Bengali image captioning tasks. Moreover, our study validates the usefulness of CLIP image encoders as feature extractors in the same domain. From our experiments, we observed that Bengali text can be encoded in multiple ways in Unicode due to variations in character composition and diacritics. To address this issue, we applied Unicode normalization, further validating our results. Due to hardware limitations, we could not experiment beyond the base GPT-2 model, which has limited parameters. Future directions include experimenting with other open-source LLMs, such as LLaMA.

REFERENCES

- [1] R. Mokady, A. Hertz, and A. H. Bermano, "Clipcap: Clip prefix for image captioning," 2021.
- [2] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," 2021.
- [3] M. Li, T. Lv, J. Chen, L. Cui, Y. Lu, D. Florencio, C. Zhang, Z. Li, and F. Wei, "Trocr: Transformer-based optical character recognition with pre-trained models," 2022.
- [4] M. Rahman, N. Mohammed, N. Mansoor, and S. Momen, "Chittrion: An automatic bangla image captioning system," 2018.
- [5] A. H. Kamal, M. A. Jishan, and N. Mansoor, "Textimage: The automated bangla caption generator based on deep learning," in *2020 International Conference on Decision Aid Sciences and Application (DASA)*, 2020, pp. 822–826.
- [6] A. M. Faruk, H. A. Faraby, M. M. Azad, M. R. Fedous, and M. K. Morol, "Image to bangla caption generation using deep cnn and bidirectional gated recurrent unit," 2020.
- [7] M. Humaira, S. Paul, M. A. R. K. Jim, A. S. Ami, and F. M. Shah, "A hybridized deep learning method for bengali image captioning," *International Journal of Advanced Computer Science and Applications*, vol. 12, no. 2, 2021. [Online]. Available: <http://dx.doi.org/10.14569/IJACSA.2021.0120287>
- [8] B. Das, R. Pal, M. Majumder, S. Phadikar, and A. A. Sekh, "A visual attention-based model for bengali image captioning," *SN Computer Science*, vol. 4, 02 2023.
- [9] R. Solomon, M. Abebe, and A. Minutolo, "Amharic language image captions generation using hybridized attention-based deep neural networks," *Appl. Comp. Intell. Soft Comput.*, vol. 2023, jan 2023. [Online]. Available: <https://doi.org/10.1155/2023/9397325>
- [10] M. A. H. Palash, M. A. A. Nasim, S. Saha, F. Afrin, R. Mallik, and S. Samiappan, "Bangla image caption generation through cnn-transformer based encoder-decoder network," in *Proceedings of International Conference on Fourth Industrial Revolution and Beyond 2021*. Singapore: Springer Nature Singapore, 2022, pp. 631–644.
- [11] F. Muhammad Shah, M. Humaira, M. A. R. K. Jim, A. Saha Ami, and S. Paul, "Bornon: Bengali image captioning with transformer-based deep learning approach," *SN Comput. Sci.*, vol. 3, no. 1, nov 2021. [Online]. Available: <https://doi.org/10.1007/s42979-021-00975-0>
- [12] N. Nawer, M. S. I. Khan, M. M. Alam, M. H. K. Mehedi, and A. A. Rasel, "Transformation of visual information into bangla textual representation," in *2023 IEEE 13th Annual Computing and Communication Workshop and Conference (CCWC)*, 2023, pp. 0436–0441.
- [13] M. Faiyaz Khan, S. M. Sadiq-Ur-Rahman, and M. Saiful Islam, "Improved bengali image captioning via deep convolutional neural network based encoder-decoder model," in *Proceedings of International Joint Conference on Advances in Computational Intelligence*. Singapore: Springer Singapore, 2021, pp. 217–229.
- [14] M. S. Salim, H. Murad, D. Das, and F. Ahmed, "Banglapgt: A generative pretrained transformer-based model for bangla language," in *2023 International Conference on Information and Communication Technology for Sustainable Development (ICT4SD)*, 2023, pp. 56–59.
- [15] M. Freitag and Y. Al-Onaizan, "Beam search strategies for neural machine translation," in *Proceedings of the First Workshop on Neural Machine Translation*. Association for Computational Linguistics, 2017. [Online]. Available: <http://dx.doi.org/10.18653/v1/W17-3207>
- [16] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," 2021.
- [17] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," 2021.
- [18] R. Vedantam, C. L. Zitnick, and D. Parikh, "Cider: Consensus-based image description evaluation," in *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 4566–4575.
- [19] M. Denkowski and A. Lavie, "Meteor universal: Language specific translation evaluation for any target language," in *Proceedings of the Ninth Workshop on Statistical Machine Translation*. Baltimore, Maryland, USA: Association for Computational Linguistics, Jun. 2014, pp. 376–380. [Online]. Available: <https://aclanthology.org/W14-3348>
- [20] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "Bleu: a method for automatic evaluation of machine translation," in *Proceedings of the 40th Annual Meeting on Association for Computational Linguistics*, ser. ACL '02. USA: Association for Computational Linguistics, 2002, p. 311–318. [Online]. Available: <https://doi.org/10.3115/1073083.1073135>
- [21] X. L. Li and P. Liang, "Prefix-tuning: Optimizing continuous prompts for generation," in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, C. Zong, F. Xia, W. Li, and R. Navigli, Eds. Online: Association for Computational Linguistics, Aug. 2021, pp. 4582–4597. [Online]. Available: <https://aclanthology.org/2021.acl-long.353>